

Module 04 Lab - Exploratory Data Analysis

Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.

This lab is fully coded. Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading

What is Exploratory Data Analysis (EDA)? EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4 # Seaborn is a library built on top of Matplotlib that makes creating beautiful plots easier.
5 # Load the dataset directly from a URL
6 df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
7 print("--- First 5 Rows ---")
8 print(df.head())
9 print("--- Basic Info ---")
10 # .info() is a great first command. It tells us the column names, how many non-null values are in each column, and the data type.
11 # Notice that 'Age' and 'Cabin' have missing values!df.info()
```

```
--- First 5 Rows ---
  PassengerId  Survived  Pclass
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

   Name                               Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris             male  22.0     1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0     1
2    Heikkinen, Miss. Laina             female  26.0     0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0     1
4    Allen, Mr. William Henry             male  35.0     0

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.25    NaN        S
1      0   PC 17599  71.28    C85        C
2      0  STON/O2. 3101282   7.92    NaN        S
3      0   113803   53.10    C123        S
4      0   373450   8.05    NaN        S

--- Basic Info ---
```

Part 2: Descriptive Statistics

Let's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
1 # Get summary statistics for numerical columns
2 print("--- Descriptive Statistics ---")
3 print(df.describe ())
4 print("--- Key Insights from Statistics ---")
5 print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
6 print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
7 print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}.")
```

```
--- Descriptive Statistics ---
   PassengerId  Survived  Pclass     Age  SibSp  \
count      891.000000   891.000000   891.000000   714.000000   891.000000
```

mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

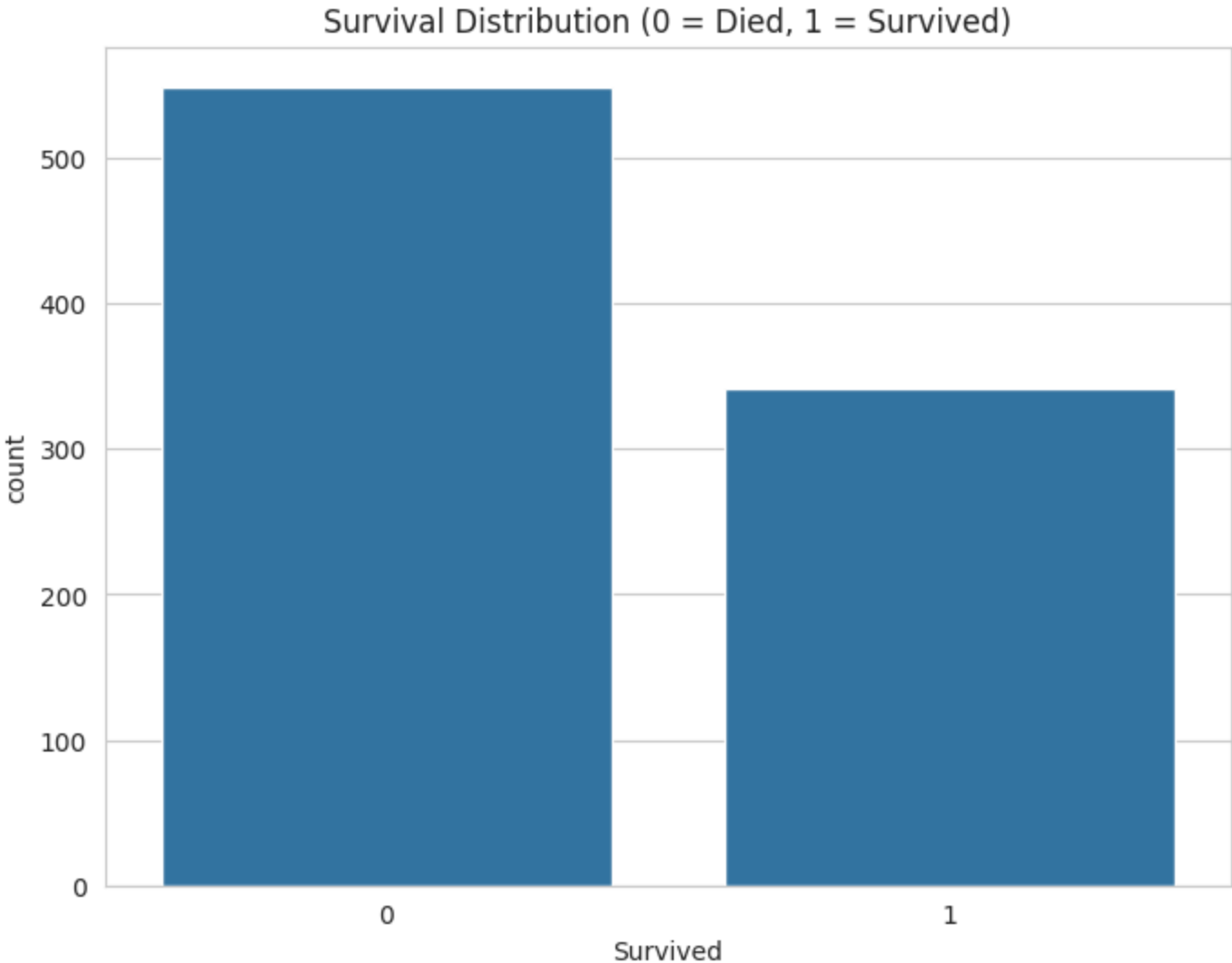
	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

--- Key Insights from Statistics ---
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from \$0.0 to a whopping \$512.3292.

Part 3: Visual EDA - Telling Stories with PlotsNumbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.**A Note on Plotting Libraries:**
* **Matplotlib:** The foundational library, gives you full control over everything.
* **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived?A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

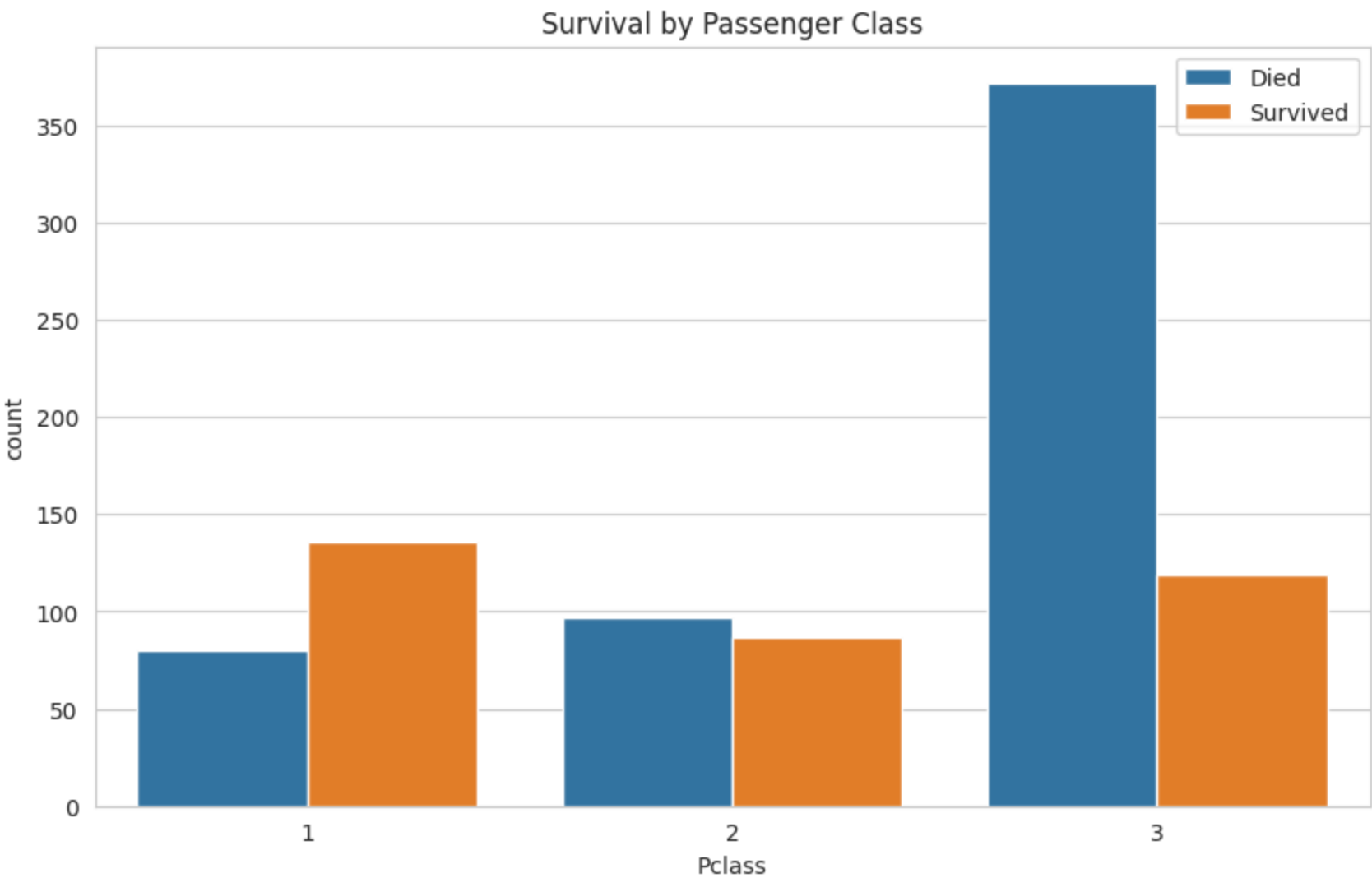
```
1 sns.set_style('whitegrid') # Sets a nice visual style for our plots
2 plt.figure(figsize=(8, 6))
3 sns.countplot(x='Survived', data=df)
4 plt.title('Survival Distribution (0 = Died, 1 = Survived)')
5 plt.show()
6 print("Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge")
```



Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge

Visualization 2: Does passenger class matter for survival?Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

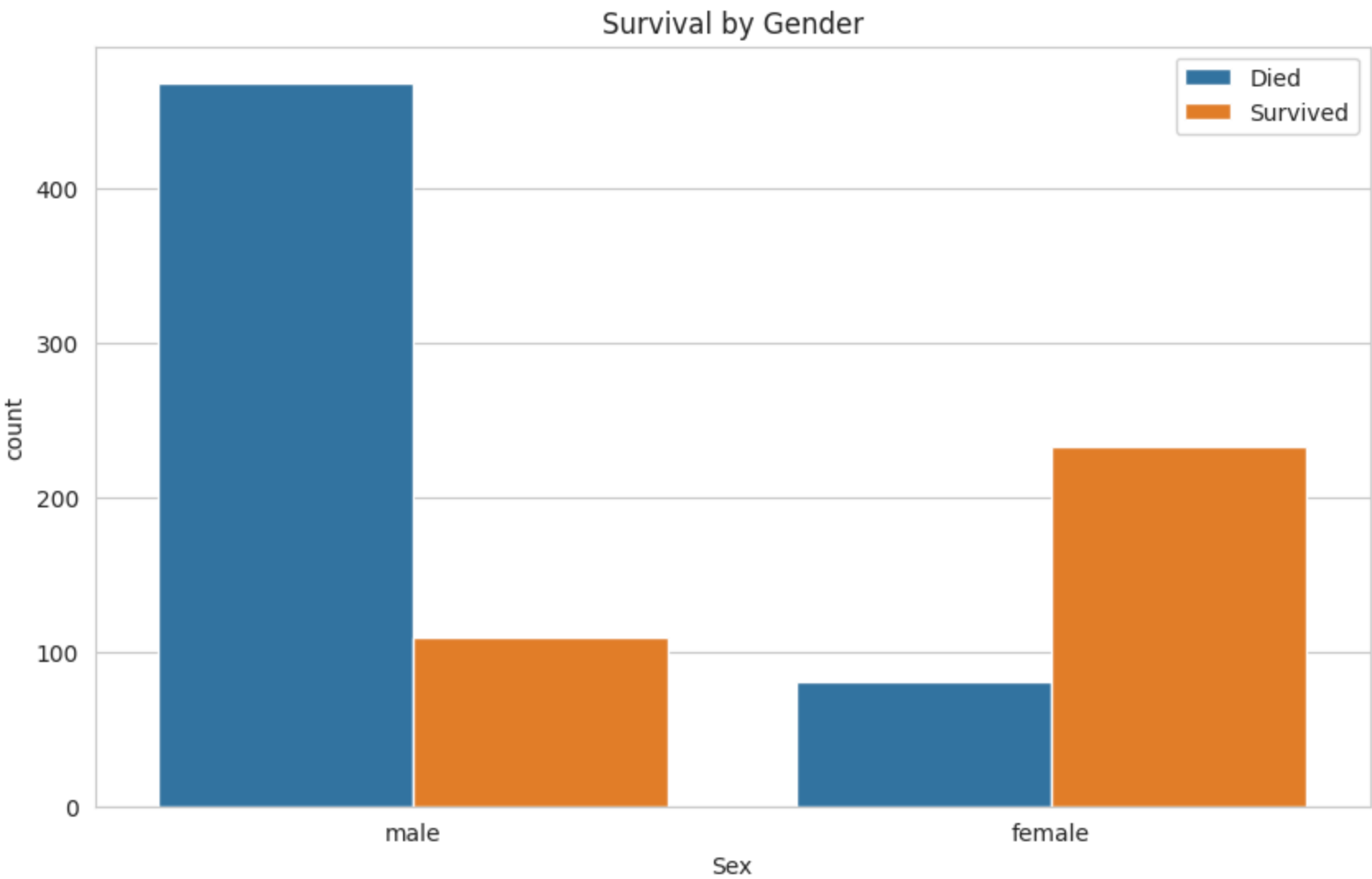
```
1 plt.figure(figsize=(10, 6))
2 sns.countplot(x='Pclass', hue='Survived', data=df)
3 plt.title('Survival by Passenger Class')
4 plt.legend(['Died', 'Survived'])
5 plt.show()
6 print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers")
```



Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers

Visualization 3: What about gender?The 'women and children first' mantra is famous. Let's see if the data supports it.

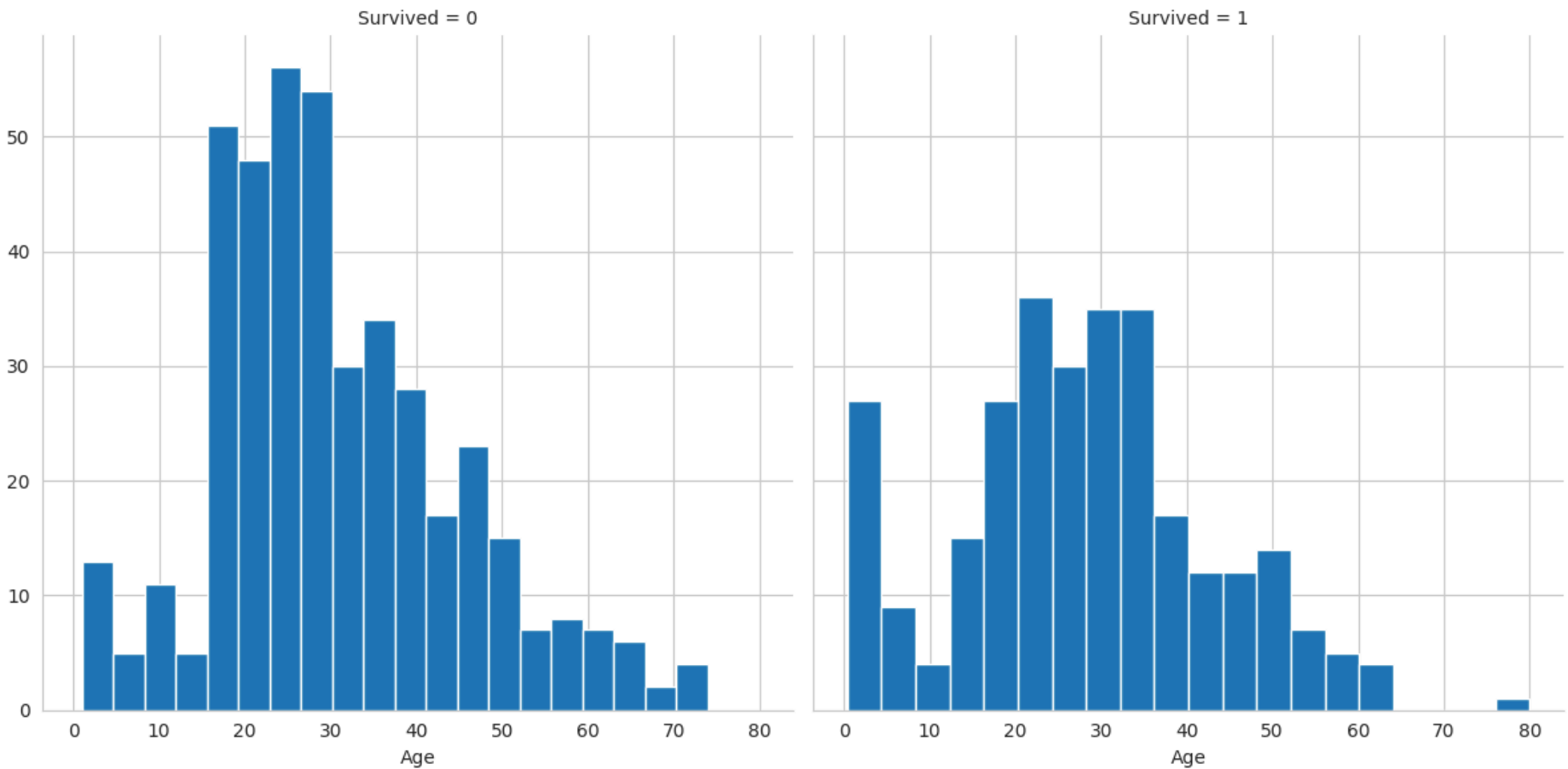
```
1 plt.figure(figsize=(10, 6))
2 sns.countplot(x='Sex', hue='Survived', data=df)
3 plt.title('Survival by Gender')
4 plt.legend(['Died', 'Survived'])
5 plt.show()
6 print("Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong pattern")
```



Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong pattern

Visualization 4: How does age play a role?For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
1 # A FacetGrid allows us to create multiple plots side-by-side to compare distributions.
2 # Here, we create one histogram for passengers who died (col='Survived'=0) and one for those who survived (col='Survived'=1)
3 g = sns.FacetGrid(df, col='Survived', height=6)
4 g.map(plt.hist, 'Age', bins=20)
5 plt.show()
6 print("Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survived, the distribution is more spread out, with a peak around 20-30 and another around 30-40.")
```

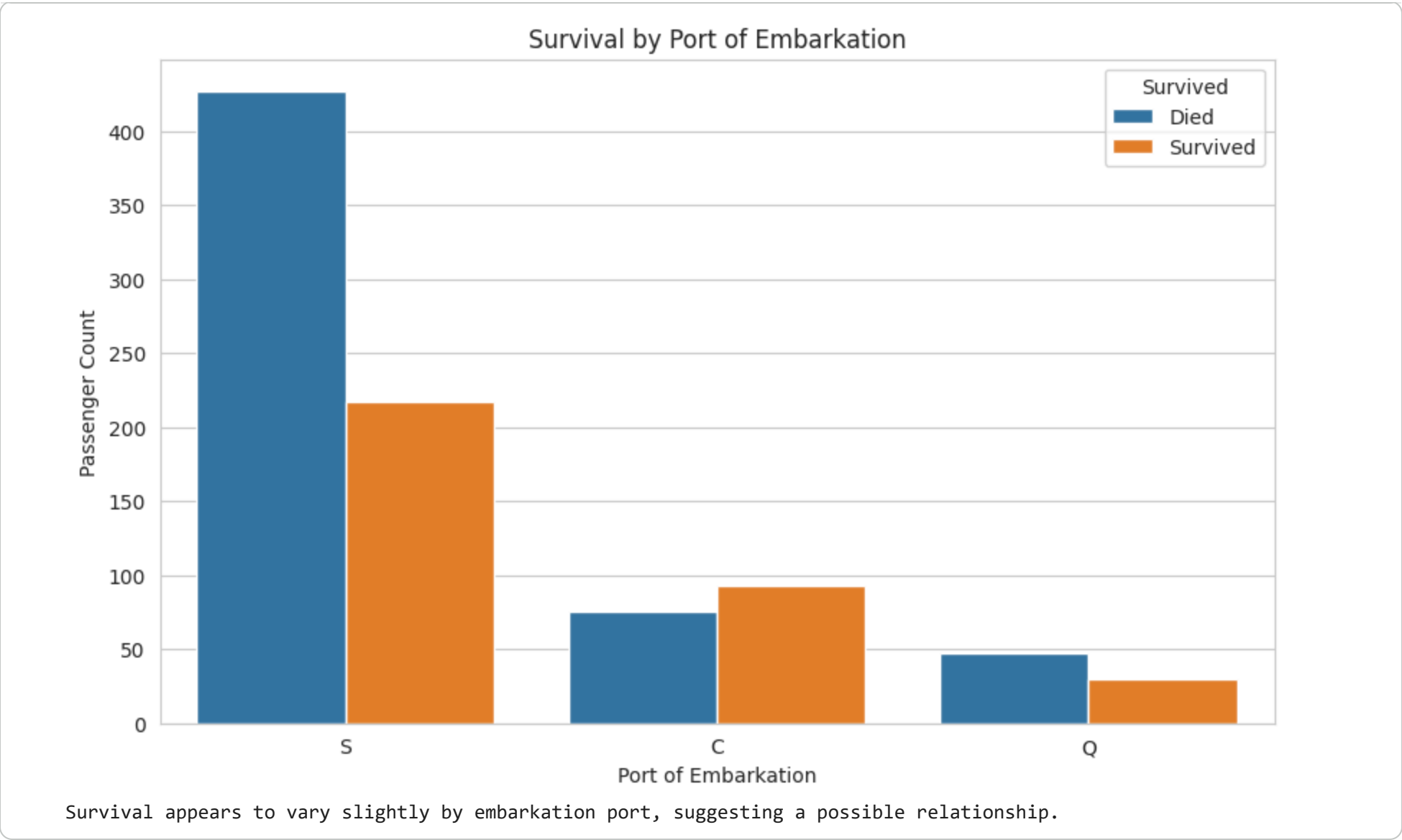


Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survive

Part 4: Student Experimentation**Instructions:** Create your own visualizations to explore other relationships in the data.

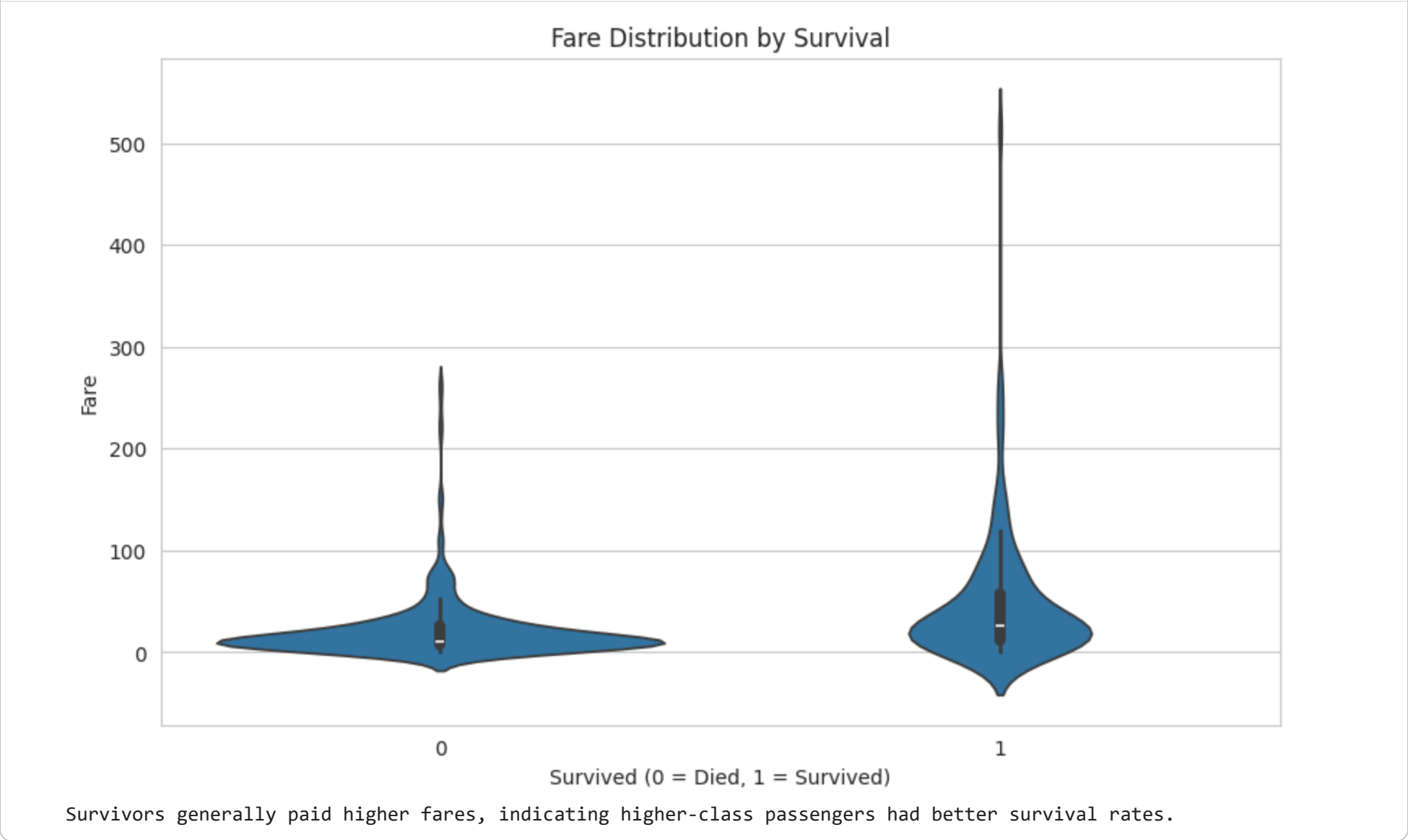
Experiment 1: Port of Embarkation1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

```
1 #Survival Rates port of Embarkation
2 plt.figure(figsize=(10, 6))
3 sns.countplot(x='Embarked', hue='Survived', data=df)
4 plt.title('Survival by Port of Embarkation')
5 plt.xlabel('Port of Embarkation')
6 plt.ylabel('Passenger Count')
7
8 plt.legend (title='Survived', labels=['Died',"Survived"])
9 plt.show()
10 print("Survival appears to vary slightly by embarkation port, suggesting a possible relationship.")
```



Experiment 2: Fare vs. Survival1. Fare is a continuous numerical variable.2. A boxplot or violinplot is a great way to see the distribution of fares for those who survived vs. those who didn't.3. Create one of these plots to compare the Fare distribution by Survived. What does it tell you?

```
1 # Fare vs. Survival
2 plt.figure(figsize=(10, 6))
3 sns.violinplot(x='Survived', y='Fare', data=df)
4
5 plt.title('Fare Distribution by Survival')
6 plt.xlabel('Survived (0 = Died, 1 = Survived)')
7 plt.ylabel('Fare')
8 plt.show()
9
10 print("Survivors generally paid higher fares, indicating higher-class passengers had better survival rates.")
```



Instructions: Answer the following questions in this markdown cell.1. ****What is the primary goal of Exploratory Data Analysis (EDA)?**

2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).

3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?

1.What is the primary goal of EDA?

The goal of EDA is to understand a dataset before building any model finding patterns, spotting missing values or errors, and seeing how variables relate to each other. It helps you catch problems early and make smarter choices when you actually build the model.

2.What kind of person had the best chance of surviving?

A young female traveling in 1st class. The plots showed females survived at a much higher rate than males, 1st class passengers survived far more often than 3rd class, and there was a noticeable spike in survival for young children.

3.Why visualize instead of just looking at summary statistics?

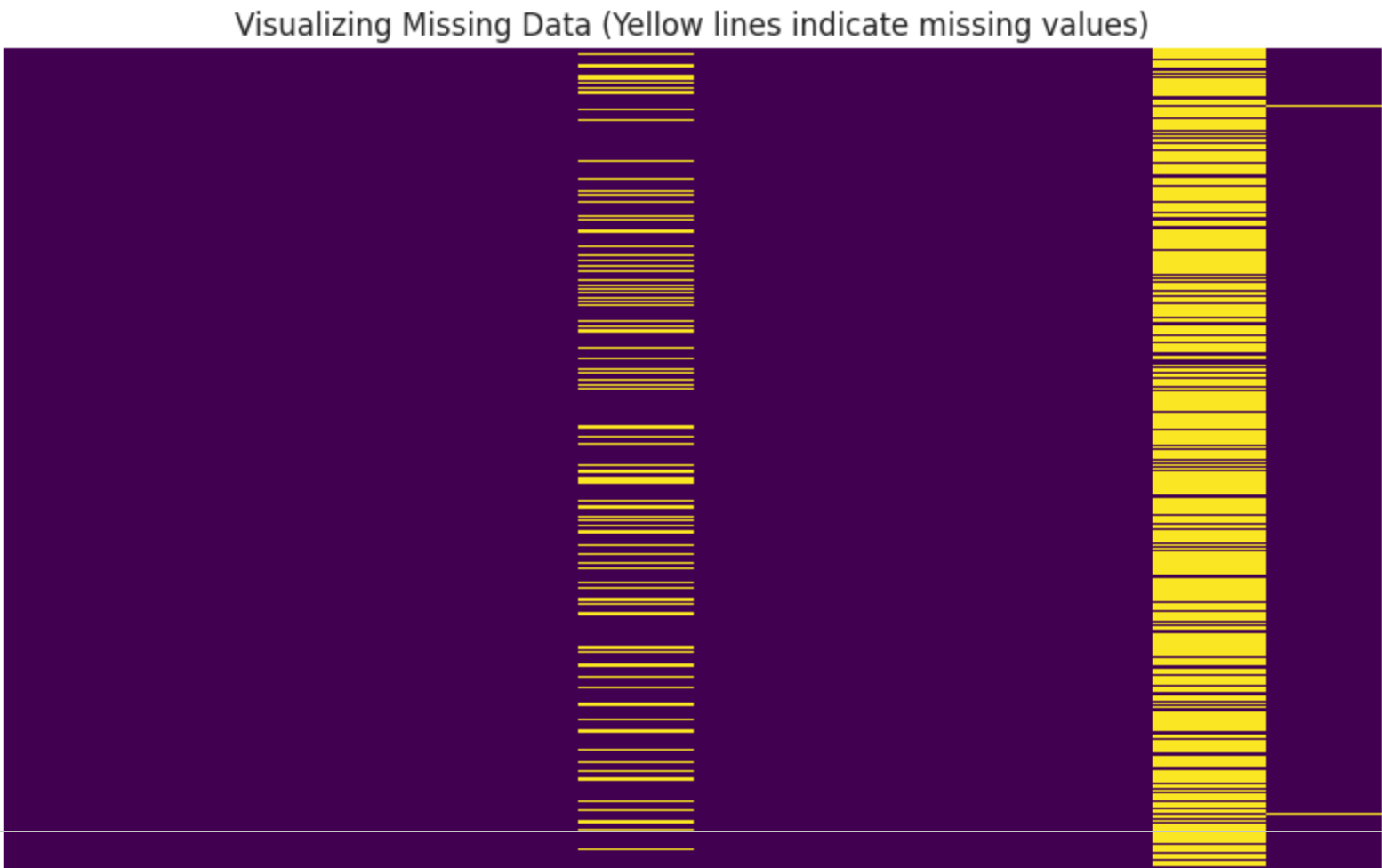
A number like the mean or a count only gives you one value and hides the bigger picture. A plot shows the shape of the data — the spread, outliers, and imbalance (like seeing that far more people died than survived), and it shows relationships between two variables at once (like survival by gender or class) that a single number can't reveal on its own.**

PERSONAL EXPERIMENT:

For this lab we will be implementing two additional excercise to understand EDA better.

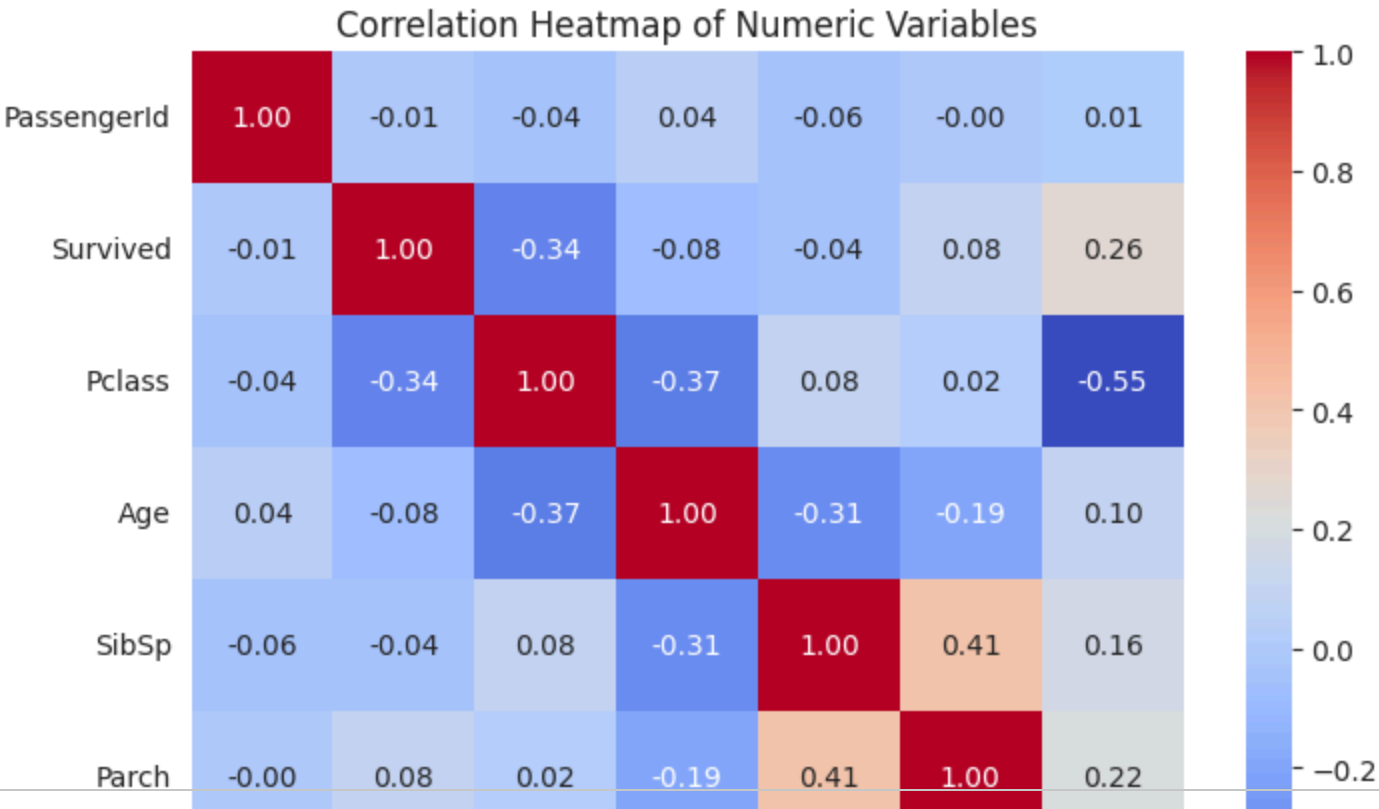
1. Missing Data Visualization: A super important part of EDA is to find out what data is missing before cleaning it. The code below indicates missing data by using yellow lines and helping us strengthen our visualization concepts.

```
1 plt.figure(figsize=(10, 6))
2 sns.heatmap(df.isnull(), yticklabels=False, cbar=False, cmap='viridis')
3 plt.title('Visualizing Missing Data (Yellow lines indicate missing values)')
4 plt.show()
```

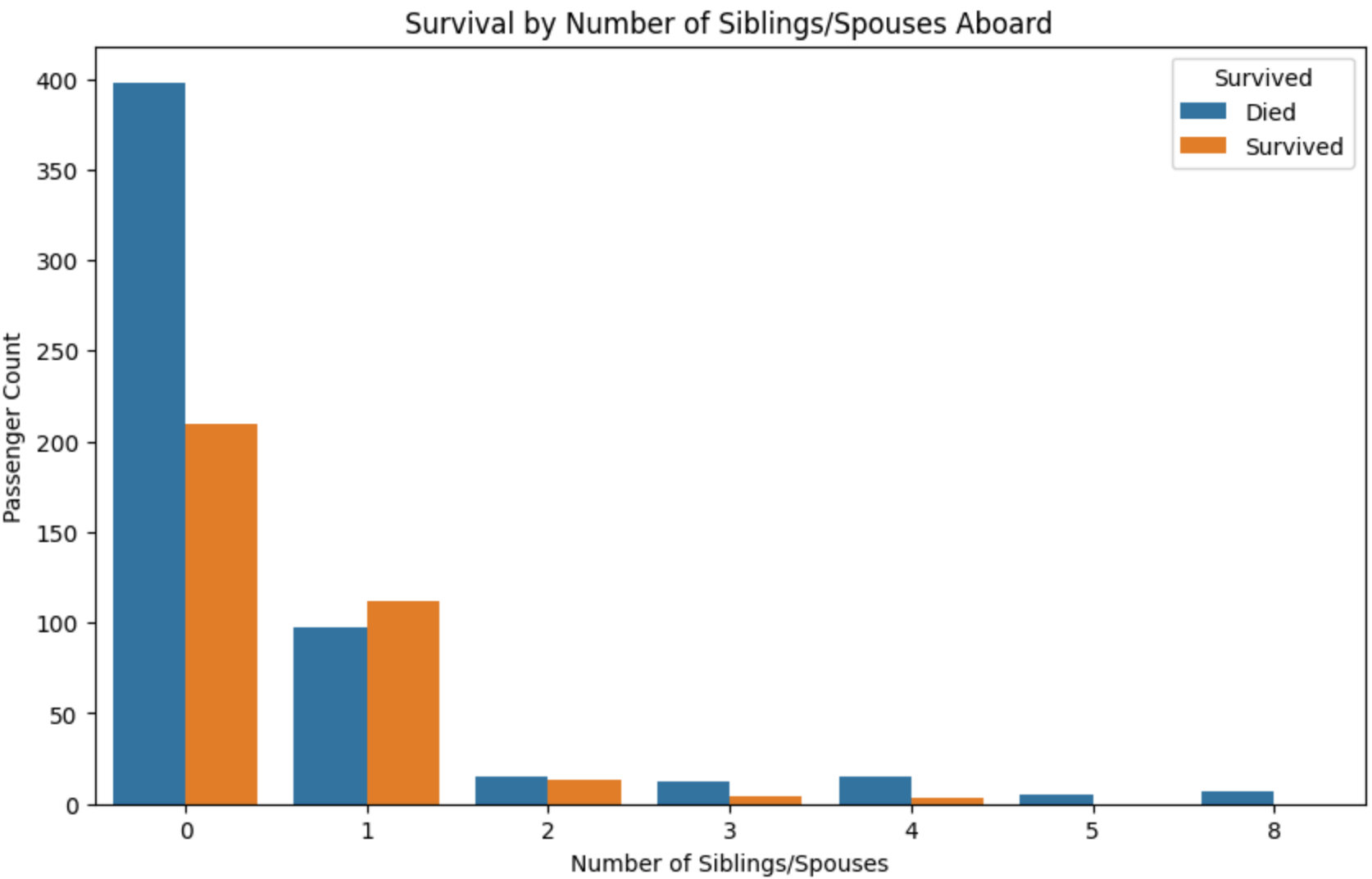


2. Correlation Heatmap: This experiment helps us understand better how all the numerical variables relate to each other at a single glance.

```
1 plt.figure(figsize=(8, 6))
2 # Select only numeric columns for correlation
3 numeric_df = df.select_dtypes(include=['number'])
4 sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
5 plt.title('Correlation Heatmap of Numeric Variables')
6 plt.show()
```



```
1 # Personal Experiment: Survival by Number of Siblings/Spouses Aboard
2 plt.figure(figsize=(10, 6))
3 sns.countplot(x='SibSp', hue='Survived', data=df)
4 plt.title('Survival by Number of Siblings/Spouses Aboard')
5 plt.xlabel('Number of Siblings/Spouses')
6 plt.ylabel('Passenger Count')
7 plt.legend(title='Survived', labels=['Died', 'Survived'])
8 plt.show()
9
10 print("Passengers traveling alone or with one sibling/spouse had better survival rates compared to those in larger fa
```



Passengers traveling alone or with one sibling/spouse had better survival rates compared to those in larger family groups